CREPE: Deep Learning for Pitch Estimation

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1 Introduction

1.1 Context and Motivation

Pitch estimation the task of recovering the fundamental frequency f_0 of an audio signal from its waveform is a foundational problem in speech and music processing, with applications ranging from speaker characterisation and prosody analysis to automatic music transcription and audio-to-MIDI conversion. Despite decades of research, robust pitch tracking remains an open challenge: classical signal processing methods exploit periodicity assumptions that break down under noise or non-stationarity, while modern deep learning approaches offer superior robustness at the cost of interpretability and computational overhead.

This report investigates and compares three representative pitch estimation methods spanning the classical-to-modern spectrum: the **Autocorrelation** method, the **YIN** algorithm [3], and **CREPE** [5], a convolutional neural network operating directly on the raw waveform. Beyond pitch estimation accuracy, we also examine a secondary but practically important task: the classification of audio frames into **voiced** and **unvoiced** segments, for which each method provides a natural decision criterion, the normalised autocorrelation peak, the CMNDF minimum, and the CREPE confidence score, respectively.

1.2 Report Structure

The remainder of this report is organised as follows. Section 2 presents the three methods from a theoretical standpoint, covering the autocorrelation function, the YIN algorithm and its cumulative mean normalised difference function, and the CREPE architecture. Section 3 describes the experimental protocol, including the noise models and evaluation metrics employed. Section 4 reports and analyses the comparative results, covering pitch stability on both synthetic and real speech data, robustness to additive and structured noise, and voiced/unvoiced classification performance. Section 5 concludes by summarising the key findings, identifying the failure modes of each approach, and outlining directions for future work.

1.3 Key Definitions

The **fundamental period** of a signal is the smallest positive time interval over which the signal repeats itself exactly. For a periodic signal $x(t)$, the fundamental period T_0 is defined such that $x(t) = x(t + T_0)$ for all t. The **fundamental frequency** f_0 is then the physical, measurable quantity defined as the reciprocal of this period:

$$f_0 = \frac{1}{T_0} \quad \text{(measured in Hz).} \tag{1}$$

While f_0 is the primary acoustic correlate of what we hear, it is distinct from **pitch**, which is a psychoacoustic attribute of auditory sensation. According to the American National Standards Institute (ANSI), pitch is "that attribute of auditory sensation in terms of which sounds may

be ordered on a scale extending from low to high" [15]. Although pitch is subjective and its perception can be influenced by factors such as timbre and loudness, it is standard practice in music information retrieval and many audio processing contexts to use the terms interchangeably, with the fundamental frequency serving as an effective physical proxy for the perceived pitch [4, 5].

1.4 Practical Considerations in Pitch Tracking

In practice, audio signals are non-stationary: the pitch changes over time. Therefore, we divide the signal into short, overlapping **analysis windows** (or frames) and estimate f_0 (the pitch) independently for each frame. This raises two fundamental questions:

1. **Window size selection**: The window must be long enough to capture several periods of the lowest expected frequency, but short enough to assume stationarity. For speech, typical window lengths range from 20–40 ms; for music, 40–60 ms is common.

2. **Fundamental frequency estimation**: How do we reliably extract f_0 from a windowed signal?

2 Methods for Pitch estimation

2.1 Autocorrelation-Based Methods

The most intuitive approach to f_0 estimation exploits the periodicity of the signal through **autocorrelation**. The autocorrelation function measures the similarity between a signal and a delayed copy of itself as a function of the lag τ:

$$r(\tau) = \sum_{n=0}^{N-1} x[n] \cdot x[n+\tau] \tag{2}$$

For a periodic signal with period T, the autocorrelation exhibits peaks at lags $\tau = 0, T, 2T, \ldots$. By finding the lag τ_{peak} corresponding to the first significant peak after zero, we obtain the period and thus the fundamental frequency: $f_0 = f_s/\tau_{\text{peak}}$, where f_s is the sampling rate. Variants of this approach include the **average magnitude difference function (AMDF)** [8] and the **normalized cross-correlation function (NCCF)** used in RAPT [9] and PRAAT [1].

2.2 The YIN Algorithm

A major advancement in autocorrelation-based pitch estimation came with the **YIN algorithm** [3], which addresses several limitations of basic autocorrelation through a series of refinements. The algorithm proceeds as follows:

1. **Difference function**: First we compute d:

$$d(\tau) = \sum_{n=1}^{N} (x[n] - x[n+\tau])^2 \tag{3}$$

   This formulation is less sensitive to amplitude variations than autocorrelation but d(τ) has a decreasing trend as τ increases. [3].

2. **Cumulative mean normalization**: Then we compute d' to compensate this decreasing trend:

$$d'(\tau) = \frac{d(\tau)}{\frac{1}{\tau}\sum_{j=1}^{\tau} d(j)} \tag{4}$$

This normalization prevents the function from favoring zero-lag ($\tau = 0$) and flattens the response, making peaks more prominent [10].

3. **Absolute thresholding**: We minimize d': the algorithm searches for the smallest τ_{min} for which $d'(\tau_{min})$ falls below a user-defined threshold (typically 0.1), indicating a candidate period [13].

4. **Parabolic interpolation**: However here our τ are discretes so we take τ_{min} and its neighboors: (τ_{min}-1, d'(τ_{min}-1)), (τ_{min}, d'(τ_{min})), and (τ_{min}+1, d'(τ_{min}+1)) We fit a parabola around these three points and then take the minimum of this parabola τ_{YIN} [13].

Applying this algorithm we get the pitch : $f_0 = \frac{f_s}{\tau_{YIN}}$.

YIN remains one of the most effective and widely cited pitch detection methods, achieving high accuracy on both speech and music [3].

2.3 CREPE: A Data-Driven Method for Pitch Estimation

2.3.1 CREPE Architecture and Training

CREPE (Convolutional Representation for Pitch Estimation) is a deep convolutional neural network that operates directly on the time-domain waveform to estimate the fundamental frequency of monophonic audio signals [5]. The model estimates pitch across a range of 32.70 Hz to 1975.5 Hz, corresponding to the spectrum of hearing of humans (20Hz-20kHz).

The architecture, shown in Figure 1, processes audio resampled at $f_s = 16$ kHz, segmented into overlapping (85%) frames of 1024 samples. This results in a frame duration of $\frac{1024}{16000} = 64$ ms, which ensures the capture of at least two periods of the lowest target frequency (32.70 Hz, period ≈ 30.6 ms). This choice of sampling rate balances temporal resolution with the ability to capture low-frequency periodicity.

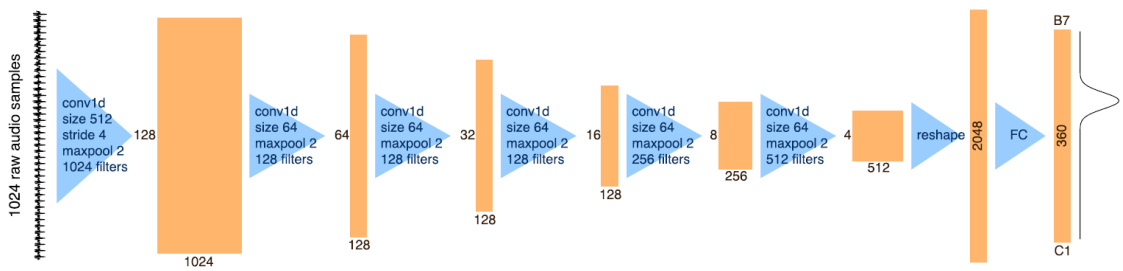

Figure 1: Architecture of the CREPE pitch tracker. The six convolutional layers operate directly on the time-domain audio signal, producing a 360-dimensional output vector that is then used to derive the pitch estimate. (Adapted from [5])

Each frame is passed through six convolutional layers, producing a 2048-dimensional latent representation. A final dense layer with sigmoid activation outputs a 360-dimensional vector $\hat{\mathbf{y}}$, where each dimension corresponds to a specific pitch value expressed in cents. The cent is a logarithmic unit representing musical intervals relative to a reference pitch $f_{\text{ref}} = 10$ Hz:

$$c(f) = 1200 \cdot \log_2 \left(\frac{f}{f_{\text{ref}}} \right),$$

The 360 output values cover frequencies from 32.70 Hz to 1975.5 Hz on a logarithmic scale, with a 20-cent interval between consecutive frequencies. The final pitch estimate $\hat{c}$ is computed as

the weighted average of the cent values c_i associated with each output unit:

$$\hat{c} = \frac{\sum_{i=1}^{360} \hat{y}_i c_i}{\sum_{i=1}^{360} \hat{y}_i}, \quad \hat{f} = f_{\text{ref}} \cdot 2^{\hat{c}/1200}.$$

During training, the target vectors are created by applying an RBF (Gaussian) kernel with a standard deviation of 25 cents to the ground truth frequency, smoothing the penalty for near-correct predictions [16]. The network then minimizes the binary cross-entropy loss between its predictions and these smoothed target vectors.

The model was trained using 5-fold cross-validation with a 60/20/20 train/validation/test split on the MDB-stem-synth dataset [17]. Artist-conditional folds were employed to prevent training and testing on the same artist, which could artificially inflate performance metrics [18].

2.3.2 Experimental Findings from the Aticle [5]

The authors report three key findings regarding CREPE's performance:

- **Accuracy:** CREPE achieves state-of-the-art raw pitch accuracy (RPA), outperforming both pYIN [6] and SWIPE [2] across standard evaluation thresholds. The performance gap widens as the evaluation tolerance decreases: at a strict 10-cent threshold, CREPE maintains over 90% accuracy on the MDB-stem-synth dataset, compared to approximately 82-83% for baseline algorithms.
- **Noise robustness:** CREPE demonstrates superior robustness to additive noise in most conditions, maintaining higher accuracy than baseline algorithms across various signal-to-noise ratios (SNR) for pub noise, white noise, and pink noise. Brown noise remains an exception where pYIN shows comparable robustness due to YIN's inherent resilience to low-frequency noise.
- **Timbre generalization:** While CREPE performs comparably to other algorithms on homogeneous datasets like RWC-synth (synthesized with fixed harmonic structure), it significantly outperforms them on the timbrally diverse MDB-stem-synth dataset, suggesting better generalization to complex, realistic timbres.

2.3.3 CREPE's Confidence Score

CREPE provides a confidence score defined as the maximum value of the output vector $\hat{\mathbf{y}}$:

$$\text{confidence} = \max_i \hat{y}_i.$$

This score reflects the model's certainty in its pitch prediction. When a clear periodic component is present in the audio, the corresponding output unit(s) exhibit high activation, yielding a high confidence score. Conversely, in noisy or ambiguous signals where no single pitch dominates, all 360 output units maintain low activation values, resulting in a low confidence score. This confidence metric can be used to identify unreliable pitch estimates or to weight pitch observations in downstream applications.

3 Experimental Protocol

3.1 Protocol

The performance of three fundamental frequency (f_0) estimation algorithms YIN, Autocorrelation, and CREPE was evaluated under varying acoustic conditions. We utilized several noise models, including Additive White Gaussian Noise (AWGN), Pink noise ($1/f$ power spectrum),

Brown noise ($1/f^2$), and non-stationary environmental noise (proxied by trumpet recordings). Robustness was quantified via the variance of the pitch estimates and the Mean Squared Error (MSE) between the noisy estimates and the clean ground truth.

3.1.1 Clean Voiced Speech Analysis

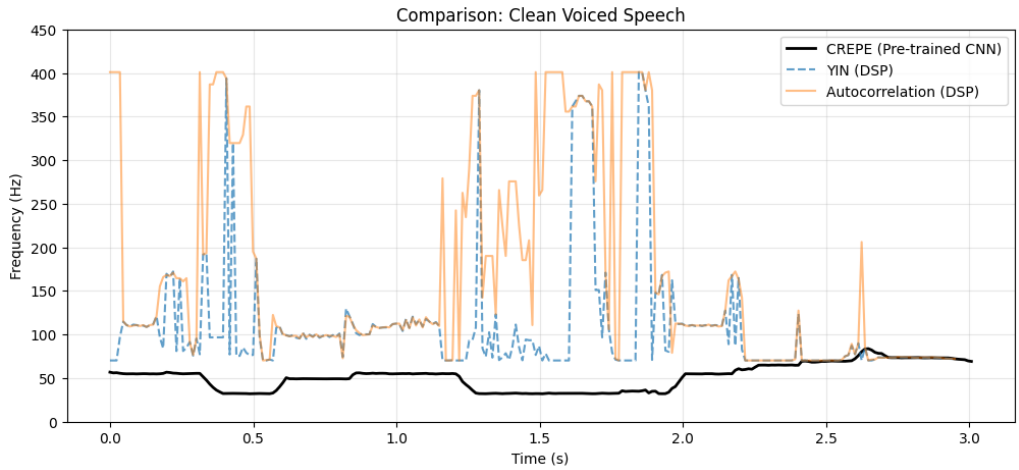

Figure 2: Pitch estimation comparison for clean voiced speech. CREPE exhibits the highest degree of temporal continuity, whereas YIN and Autocorrelation demonstrate localized instabilities.

In the clean voiced speech condition, CREPE provides a significantly smoother trajectory. In contrast, YIN exhibits intermittent instabilities, while the Autocorrelation method displays significant variance, suggesting a lower inherent precision in fundamental frequency tracking even in noiseless environments.

3.1.2 Unvoiced Speech Segments

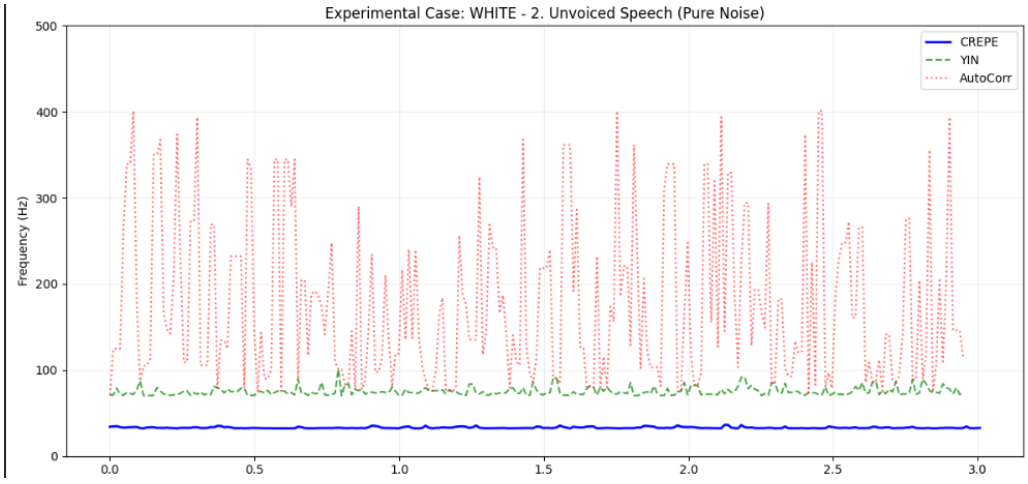

Figure 3: Algorithm behavior on unvoiced (stochastic) segments. Both CREPE and YIN maintain relatively stable, though arbitrary, estimates compared to the erratic behavior of Autocorrelation.

Analysis of unvoiced segments reveals that while no true fundamental frequency exists, the algorithms react differently to stochastic input. CREPE and YIN maintain a relatively low-

variance output, whereas the Autocorrelation method produces an erratic signal characterized by extreme variance, reflecting its sensitivity to non-periodic spectral components.

3.1.3 Robustness to Degraded Signals

To evaluate algorithmic robustness, noise was added at Signal-to-Noise Ratio (SNR) levels of 20 dB, 10 dB, and 0 dB.

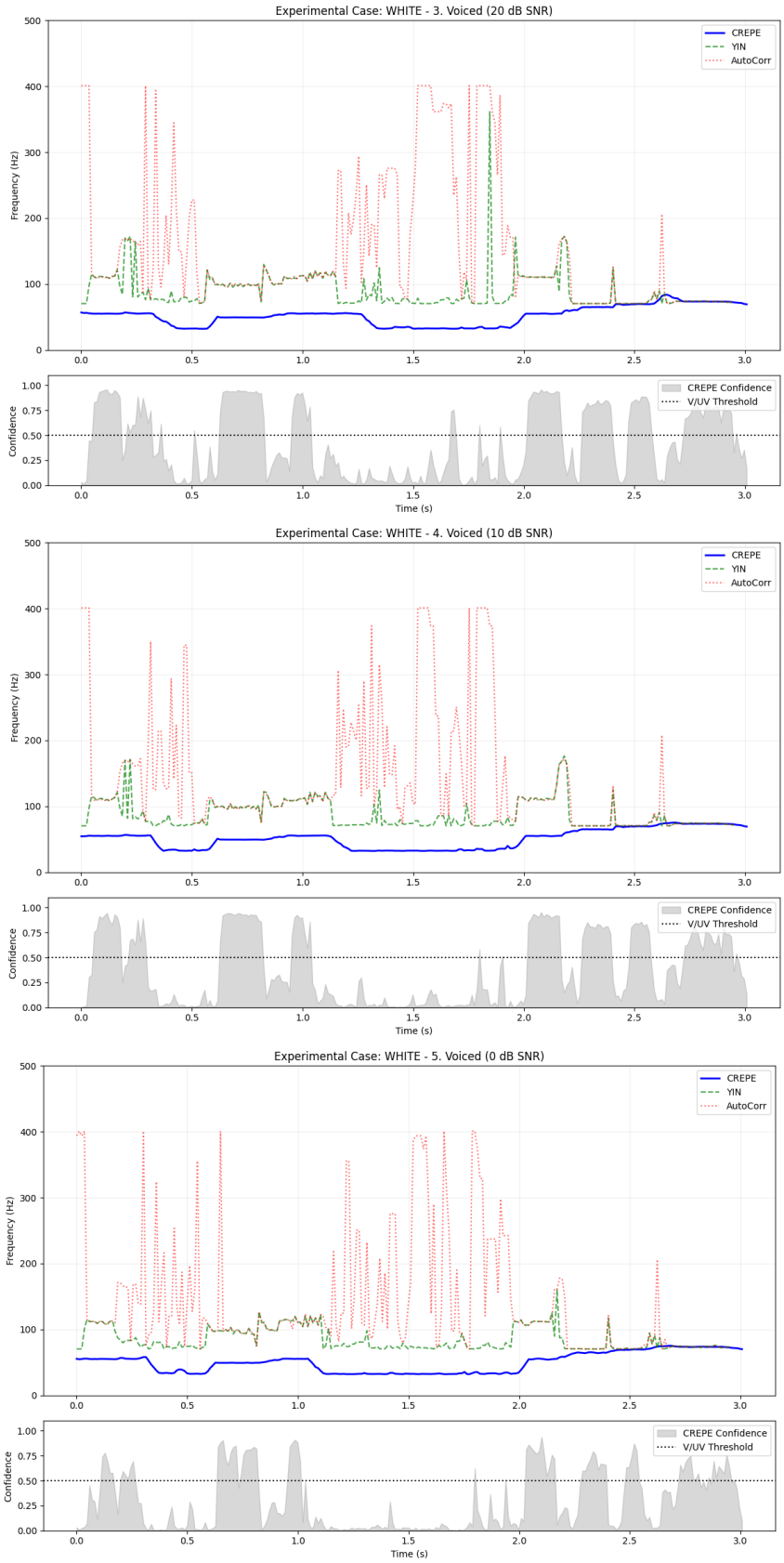

Figure 4: Evolution of pitch estimation across decreasing SNR levels (White Noise). Note the emergence of spurious peaks and discontinuities in YIN and Autocorrelation estimates as noise intensity increases.

The experimental data indicates that as the SNR decreases, the structural integrity of the pitch contours for YIN and Autocorrelation degrades significantly. These methods exhibit high sensitivity to additive noise, resulting in errors and discontinuities, which translate to a high MSE. CREPE demonstrates superior robustness, maintaining contour consistency even at 0 dB SNR, though MSE increases as the model eventually tracks noise-induced harmonics.

3.2 Comparative Statistical Performance

3.2.1 White Gaussian Noise (WGN)

White noise represents a purely random signal with equal intensity across all frequencies, similar to the static "hiss" of an untuned radio.

CREPE remains exceptionally robust, maintaining low MSE even at 0dB. YIN and Autocorrelation collapse under higher noise, with YIN's MSE exceeding 5500 at 0dB, indicating a complete loss of tracking accuracy compared to CREPE's minimal error.

Table 1: Comparative Metrics: White Noise

Condition	CREPE	YIN	AutoCorr
Speech	$\sigma^2 = 215.79$	$\sigma^2 = 4709.97$	$\sigma^2 = 13407.15$
Unvoiced	$\sigma^2 = 5.89$	$\sigma^2 = 39.31$	$\sigma^2 = 7723.03$
20dB	$\sigma^2 = 178.89$ MSE=33.01	$\sigma^2 = 1126.44$ MSE=4236.03	$\sigma^2 = 9923.79$ MSE=5413.11
10dB	$\sigma^2 = 198.05$ MSE=50.74	$\sigma^2 = 447.84$ MSE=5313.56	$\sigma^2 = 10265.85$ MSE=6680.64
0dB	$\sigma^2 = 207.08$ MSE=14.56	$\sigma^2 = 323.55$ MSE=5568.77	$\sigma^2 = 8298.08$ MSE=8925.16

3.2.2 Pink Noise

Pink noise ($1/f$) has more energy at lower frequencies, resembling natural sounds like rainfall or wind.

While all methods perform well at 20dB, CREPE experiences a significant failure at 0dB ($\sigma^2 \approx 4660$), likely due to noise energy interfering with the model's neural layers. Autocorrelation remains highly unstable throughout, with MSE values nearly triple those of YIN.

Table 2: Comparative Metrics: Pink Noise

Condition	CREPE	YIN	AutoCorr
Speech	$\sigma^2 = 215.79$	$\sigma^2 = 4709.97$	$\sigma^2 = 13407.15$
Unvoiced	$\sigma^2 = 5.89$	$\sigma^2 = 39.31$	$\sigma^2 = 7723.03$
20dB	$\sigma^2 = 226.44$ MSE=9.85	$\sigma^2 = 1877.24$ MSE=5006.57	$\sigma^2 = 15750.26$ MSE=6120.29
10dB	$\sigma^2 = 233.00$ MSE=33.13	$\sigma^2 = 1244.51$ MSE=4343.76	$\sigma^2 = 15894.62$ MSE=10533.41
0dB	$\sigma^2 = 4660.91$ MSE=4052.52	$\sigma^2 = 1787.27$ MSE=4987.64	$\sigma^2 = 19682.93$ MSE=15760.88

3.2.3 Brown Noise

Brown noise ($1/f^2$) is heavily weighted toward low frequencies, sounding like a deep rumble or distant thunder.

CREPE shows near-perfect stability (σ^2 stable around 200), as the low-frequency noise does not mask the higher-pitched speech components. YIN and Autocorrelation show massive variance ($\sigma^2 > 8000$ and > 20000 respectively at 0dB), proving their inability to distinguish the rumble from the pitch.

Table 3: Comparative Metrics: Brown Noise

Condition	CREPE	YIN	AutoCorr
Speech	$\sigma^2 = 215.79$	$\sigma^2 = 4709.97$	$\sigma^2 = 13407.15$
Unvoiced	$\sigma^2 = 5.89$	$\sigma^2 = 39.31$	$\sigma^2 = 7723.03$
20dB	$\sigma^2 = 215.72$ MSE=1.22	$\sigma^2 = 6322.37$ MSE=3972.51	$\sigma^2 = 15027.50$ MSE=2200.42
10dB	$\sigma^2 = 203.00$ MSE=67.14	$\sigma^2 = 6519.79$ MSE=4459.21	$\sigma^2 = 17813.80$ MSE=5587.08
0dB	$\sigma^2 = 234.82$ MSE=146.89	$\sigma^2 = 8846.89$ MSE=6564.18	$\sigma^2 = 21965.80$ MSE=18926.91

3.2.4 Environmental (Trumpet) Noise

This represents non-stationary noise with harmonic content, simulating real-life interference like a background musical instrument.

CREPE suffers a catastrophic breakdown at 10dB (MSE $> 110,000$), likely because the trumpet's harmonics confuse the neural network. Ironically, Autocorrelation shows lower MSE here than in other noise types, as it may be partially locking onto the trumpet's pitch rather than falling into random noise.

Table 4: Comparative Metrics: Trumpet Noise

Condition	CREPE	YIN	AutoCorr
Speech	$\sigma^2 = 215.79$	$\sigma^2 = 4709.97$	$\sigma^2 = 13407.15$
Unvoiced	$\sigma^2 = 5.89$	$\sigma^2 = 39.31$	$\sigma^2 = 7723.03$
20dB	$\sigma^2 = 200.51$ MSE=50.72	$\sigma^2 = 5405.48$ MSE=2174.56	$\sigma^2 = 10977.96$ MSE=2261.46
10dB	$\sigma^2 = 42639.08$ MSE=110246.12	$\sigma^2 = 7038.04$ MSE=7336.35	$\sigma^2 = 10771.70$ MSE=3250.37
0dB	$\sigma^2 = 7666.33$ MSE=167253.54	$\sigma^2 = 7677.25$ MSE=9441.61	$\sigma^2 = 10657.86$ MSE=7562.30

The results demonstrate that while **CREPE** provides superior robustness against stochastic interference (White, Pink, and Brown noise), its performance degrades significantly in the presence of structured, tonal interference. Because CREPE and traditional DSP methods are designed to estimate the fundamental frequency of a single source, non-stationary noise with its own harmonic content such as the **Trumpet** model introduces a competing pitch. In these scenarios, the model fails to differentiate between the speech and the interference, leading to tracking errors and extreme MSE values as it attempts to resolve two simultaneous sources.

3.3 Quantitative Evaluation on Synthetic Data

The performance of the tracking algorithms was assessed using Raw Pitch Accuracy (RPA) across three cent-based tolerance thresholds. Synthetic tonal signals provide an ideal benchmark as they eliminate environmental noise and recording artifacts, allowing for a direct measurement of algorithmic precision.

Table 5: Raw Pitch Accuracy (RPA) at Variable Cent Thresholds

Threshold	CREPE	YIN	AutoCorr
10 cents	99.20%	79.60%	71.60%
25 cents	100.00%	99.60%	99.60%
50 cents	100.00%	100.00%	99.60%

The results demonstrate that CREPE achieves near-perfect accuracy, reaching an RPA of 100.00% at the 25 and 50-cent thresholds. Most notably, at the strictest tolerance of 10 cents, CREPE maintains a high precision of 99.20%, significantly outperforming YIN (79.60%) and Autocorrelation (71.60%). This disparity at lower thresholds underscores CREPE's superior fine-pitch resolution, whereas the DSP-based methods exhibit higher frequency jitter even when tracking structurally simple synthetic signals.

4 Comparison between CREPE (deep learning method) and DSP methods (autocorrelation and YIN method)

4.1 Pitch Stability

4.1.1 Synthetic data

To systematically evaluate the temporal response characteristics of each pitch estimation algorithm, a synthetic test signal was constructed comprising alternating voiced and unvoiced segments. The stimulus consisted of oscillating pitch contours a with 1.5 second simulated voiced segments interleaved then 1 second pauses containing silence (unvoiced regions) and then simulating voiced segment.

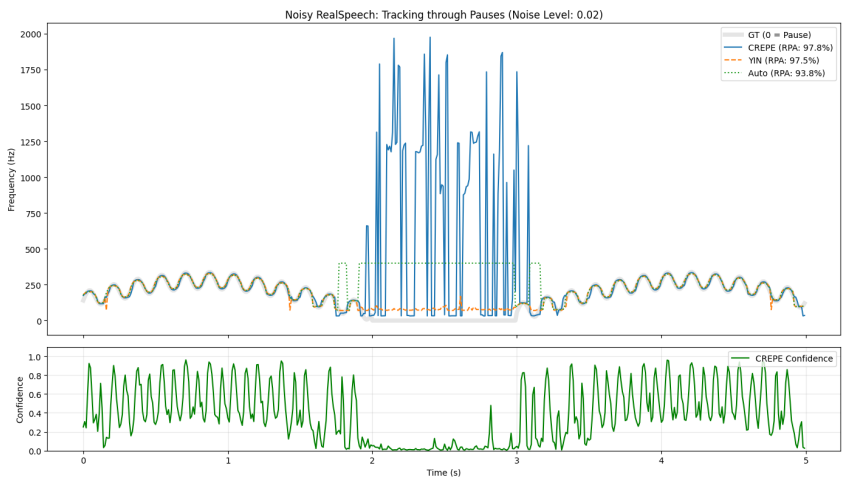

Figure 5: Comparative pitch tracking behavior across voiced and unvoiced regions. CREPE demonstrates enhanced stability during voiced segments but exhibits increased variability in the presence of white noise relative to YIN.

Table 6: Raw Pitch Accuracy (RPA) by Algorithm on Voiced Segments

Algorithm	**RPA (%)**
CREPE	97.8
YIN	97.5
Autocorrelation	93.8

Note: RPA is computed exclusively on voiced segments to isolate pitch estimation accuracy from voicing detection performance.

Behavior During Voiced Regions: All three algorithms CREPE, YIN, and autocorrelation successfully tracked the synthetic pitch contour during sustained voicing, with CREPE exhibiting marginally superior temporal stability and a better RPA. This aligns with CREPE's design objective of learning smooth pitch representations from training data [5].

Behavior During Unvoiced Regions: In unvoiced intervals (white noise), YIN demonstrated the most stable output, maintaining relatively consistent although arbitrary pitch estimates.Autocorrelation exhibited moderate instability, while CREPE produced highly erratic estimates characterized by extreme variance. However we can see that the confidence of predictions for CREPE remains very low.

Transition Dynamics at Voiced Onset: A consistent temporal lag was observed when algorithms tried to predict pitch after and before the 1 second pause (white noise), with differential recovery times reflecting each method's architecture. YIN and CREPE required approximately 100 ms to re-establish stable pitch tracking following the pause, while autocorrelation needed roughly 200 ms to recover.

This latency stems from window contamination: at voiced onset, the analysis window contains residual noise from the preceding pause. Autocorrelation-based methods require sufficient periodic content for peak detection [3]; aperiodic components attenuate the normalized peak, delaying threshold crossing. YIN inherits this through its absolute thresholding [3], where residual noise elevates the difference (d') function until periodic samples dominate.

CREPE's 1024 sample window (64 ms at 16 kHz) [5] places approximately 32 ms of preceding noise within its convolutional receptive field at onset, suppressing harmonic activation. For 1-2 frames the predictions are unstable (approximately 100 ms) until voiced content predominates.

These differential recovery times reflect each algorithm's noise robustness rather than a correctable deficiency, representing the intrinsic trade off between temporal resolution and periodicity detection in short-time analysis (as explained in the second paragraph of 2.3.1).

4.1.2 Sustained Voiced Segments

As observed in Figures 2 and 5, CREPE exhibits superior stability on sustained voiced segments compared to YIN, which in turn outperforms autocorrelation. This hierarchical stability is quantitatively confirmed in Tables 1, 2, 3, and 4, where CREPE consistently achieves the lowest variance in pitch predictions across all noise conditions.

For clean speech (Table 1), CREPE demonstrates markedly lower variance ($\sigma^2 = 215.79$) compared to YIN ($\sigma^2 = 4709.97$) and autocorrelation ($\sigma^2 = 13407.15$). This pattern persists across all noise types and SNR levels, with CREPE maintaining variance approximately an order of magnitude lower than the DSP-based methods under stochastic noise conditions.

These findings align with the synthetic data analysis in the previous part, where CREPE achieved the highest Raw Pitch Accuracy (97.8%) on simulated voiced segments, marginally outperforming YIN (97.5%) and substantially exceeding autocorrelation (93.8%). The consistency

between real speech recordings and controlled synthetic stimuli confirms that CREPE's superior stability on voiced material is a robust characteristic of the model rather than an artifact of specific test conditions.

We can therefore conclude that for sustained voiced segments, CREPE constitutes the most stable pitch estimation method among the three algorithms evaluated..

4.2 Voiced/Unvoiced segment classification : Thresholding and Decision Logic

4.2.1 Dataset

The evaluation utilizes the TEN VAD dataset [19], a publicly available resource designed for benchmarking voice activity detection systems. The dataset comprises 25 sustained voiced segments, each several seconds in duration. A key characteristic of this dataset is its provision of frame-level annotations indicating speech presence, which were precisely manually labeled. As the voiced part of audio represents 70% of the dataset, we add white noise at the end of each audio to balance the dataset.

4.2.2 Voiced/unvoiced prediction with CREPE

Threshold-based Classification A threshold-based decision rule was applied to CREPE's confidence score for voiced/unvoiced classification. For each audio sample, the optimal threshold maximizing the F1-score was selected, and these values were subsequently averaged to obtain a global decision threshold of $\eta = 0.091$. Applying this threshold yielded an accuracy of 0.741 and an F1-score of 0.784, with precision $P = 0.672$ and recall $R = 0.941$. The corresponding confusion matrix is:

$$\text{CM} = \begin{bmatrix} 136 & 116 \\ 15 & 238 \end{bmatrix} \tag{5}$$

where rows represent ground truth (unvoiced top, voiced bottom) and columns represent predictions (unvoiced left, voiced right). The high recall indicates that CREPE successfully identifies 94.1% of voiced frames, though the moderate precision (67.2%) suggests a tendency to misclassify unvoiced frames as voiced, resulting in 116 false positives.

MLP-based Classification To potentially improve upon simple thresholding, we trained a small Multi-Layer Perceptron (MLP) to classify voiced/unvoiced frames using CREPE's confidence scores. The architecture consists of three linear layers with ReLU activations, dropout for regularization, and a sigmoid output:

- Input layer: 1-dimensional (confidence value)
- Hidden layer 1: 32 units, ReLU, dropout 0.2
- Hidden layer 2: 16 units, ReLU, dropout 0.1
- Output layer: 1 unit with sigmoid activation

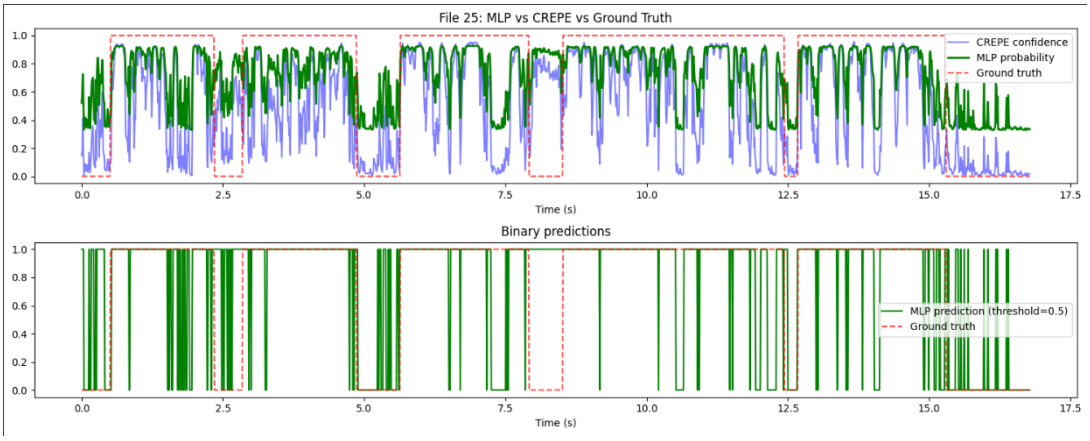

Figure 6: Classification using a simple MLP on raw confidence scores.

We then extended the feature space by incorporating temporal context: for each frame, we computed statistical features (mean, standard deviation, maximum, minimum) over a ± 20 frame window surrounding the current frame, resulting in a 5-dimensional input vector.

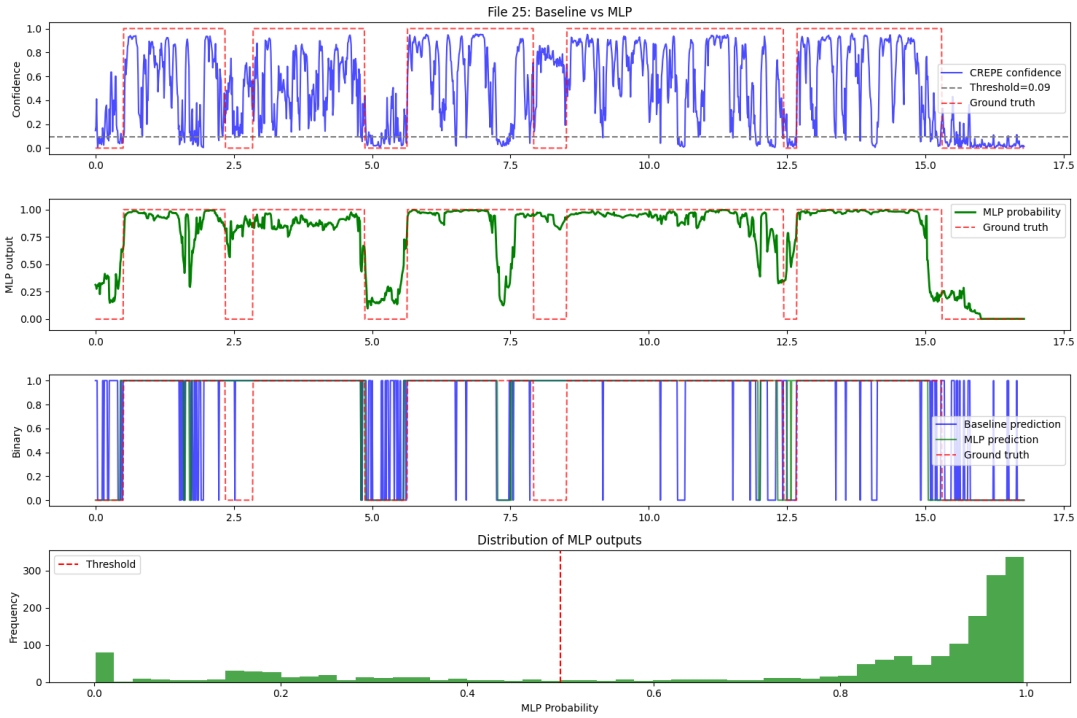

Figure 7: Classification using an MLP with temporal features (mean, max, std, min over local window).

Table 7: Comparative Performance: Threshold Baseline vs. MLP Approaches

File	Metric	Threshold	Simple MLP	MLP + Features
File 23	Accuracy	0.782	0.738	0.782
	F1-score	0.849	0.807	0.844
	Precision	0.896	-	0.765
	Recall	0.670	-	0.942
File 24	Accuracy	0.748	0.762	0.748
	F1-score	0.786	0.775	0.767
	Precision	-	-	0.896
	Recall	-	-	0.670
File 25	Accuracy	0.803	0.870	0.805
	F1-score	0.874	0.916	0.872
	Precision	-	-	0.867
	Recall	-	-	0.877

The results indicate that while the feature-enhanced MLP achieves high recall (94.2% on File 23), it struggles with precision on short pauses, which constitute the primary source of classification errors. The simple threshold baseline ultimately yields more consistent performance across files, suggesting that for this specific task, the confidence score itself provides a sufficiently discriminative feature without requiring additional temporal context or model complexity.

4.2.3 Voiced/unvoiced prediction with Autocorrelation

4.2.4 Autocorrelation Peak Picking

Voiced/unvoiced classification in autocorrelation-based pitch tracking is achieved by thresholding the normalized peak magnitude. Following the procedure described in Section 2.1, the candidate period $\hat{\tau}$ is selected as:

$$\hat{\tau} = \underset{\tau \in [\tau_{\min}, \tau_{\max}]}{\arg\max} \frac{r(\tau)}{r(0)} \tag{6}$$

A frame is then classified as voiced if the normalized peak exceeds a predefined threshold:

$$\text{Decision} = \begin{cases} \text{Voiced} & \text{if } \dfrac{r(\hat{\tau})}{r(0)} > \eta_{\text{ac}} \\ \text{Unvoiced} & \text{otherwise} \end{cases} \tag{7}$$

The threshold parameter is selected doing a grid-search.

4.2.5 YIN Peak Picking for Voiced/Unvoiced Classification

Voiced/unvoiced classification in YIN-based pitch tracking is achieved by thresholding the minimum of the Cumulative Mean Normalised Difference Function (CMNDF). Following the procedure described in Section 2.2, the CMNDF is computed over the search range $[\tau_{\min}, \tau_{\max}]$, where $\tau_{\min} = \lfloor f_s / f_{\max} \rfloor$ and $\tau_{\max} = \lfloor f_s / f_{\min} \rfloor$, with $f_{\min} = 70\,\text{Hz}$ and $f_{\max} = 400\,\text{Hz}$ constraining the search to the physiological range of human speech.

The voicing decision is based on the global minimum of $d'(\tau)$ within the pitch search range:

$$\hat{d}' = \min_{\tau \in [\tau_{\min},\, \tau_{\max}]} d'(\tau) \tag{8}$$

A frame is then classified as voiced if this minimum falls below a predefined threshold:

$$\text{Decision} = \begin{cases} \text{Voiced} & \text{if } \hat{d}' < \eta_{\text{yin}} \\ \\ \text{Unvoiced} & \text{otherwise} \end{cases} \tag{9}$$

where $d'(\tau)$ takes values in $[0, 1]$ by construction: a value near 0 indicates strong periodicity at lag τ (voiced frame), while a value near 1 indicates an aperiodic frame (unvoiced).

A silence gate is applied prior to the CMNDF computation: any frame whose peak absolute amplitude falls below 10^{-4} is directly assigned to the unvoiced class without further processing, avoiding numerical instability on silent segments.

The threshold η_{yin} is selected following a grid-search.

4.2.6 Comparative Evaluation of Voiced/Unvoiced Classification

We evaluated the three pitch estimation algorithms on the binary classification task of discriminating voiced from unvoiced frames. For each method, we selected the optimal decision threshold by maximizing the F1-score on a per-file basis, then averaged these thresholds to obtain a global decision rule or a grid-search.

Table 8: Voiced/Unvoiced Classification Performance

Metric	**CREPE**	**Autocorrelation**	**YIN**
Optimal Threshold η	0.092	0.152	0.424
Accuracy	0.8136	**0.8375**	0.8147
F1-Score	0.8184	**0.8505**	0.8095
Confusion Matrices			
	$\begin{bmatrix} 13431 & 4238 \\ 2120 & 14327 \end{bmatrix}$	$\begin{bmatrix} 12808 & 4861 \\ 682 & 15765 \end{bmatrix}$	$\begin{bmatrix} 14356 & 3313 \\ 3010 & 13437 \end{bmatrix}$

Counterintuitively, the autocorrelation method achieves the highest classification performance, with an accuracy of 83.75% and an F1-score of 0.8505. This result is noteworthy given the earlier observations of autocorrelation's instability for pitch estimation on sustained voiced segments (Section 4.2.2). The confusion matrices reveal the underlying trade-offs: autocorrelation misclassifies only 682 voiced frames as unvoised (false negatives), indicating exceptional sensitivity to periodic content, at the cost of 4861 false positives on unvoiced regions.

YIN exhibits the most conservative behavior, with the highest threshold ($\eta = 0.424$) and the fewest false positives (3313), but also the highest false negative rate (3010). CREPE strikes an intermediate balance, with a false positive count (4238) between the two DSP methods.

These results suggest that while autocorrelation proves unreliable for precise pitch tracking, its binary decision function based solely on the presence of a sufficiently strong peak serves as an effective voiced/unvoiced detector. The method's simplicity appears advantageous for this specific classification task, where the goal is periodicity detection rather than accurate frequency estimation.

4.3 Failure Cases

4.3.1 Pitch Estimation Errors

As detailed in 4.1 (Figure 5), the primary limitation across all algorithms lies in their temporal response to transitions between voiced and unvoiced regimes. When successive voiced segments

are separated by less than approximately 100 ms of silence, the algorithms fail to resolve the intervening unvoiced region, erroneously producing a continuous pitch contour. This phenomenon is evident in Figure 5 between 2.0 s and 3.0 s, where the 1 s pause should yield zero pitch, yet all methods exhibit unstable activity during the transition windows.

Under white noise conditions, CREPE demonstrates marked instability characterized by erratic pitch estimates accompanied by low confidence scores. This behavior stems from a distribution shift between training and inference: CREPE was trained on clean speech signals [5] and lacks exposure to purely stochastic inputs, causing its convolutional filters to activate on noise-induced spurious correlations.

Lastly in section 3 we have seen that CREPE yields very spurious pitch results when there are two sources at the same time.

4.3.2 Classification Errors

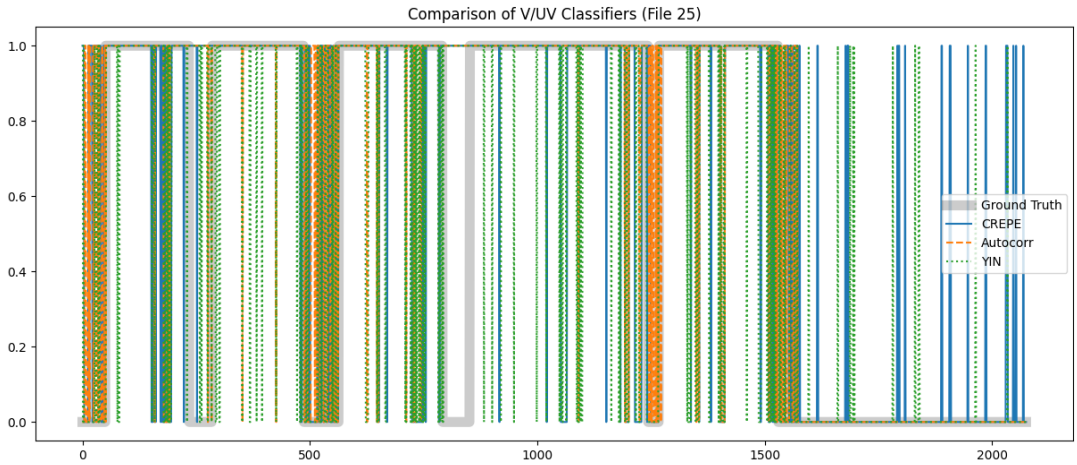

Figure 8: Temporal evolution of voiced/unvoiced classification errors across algorithms. False positives (unvoiced frames misclassified as voiced) constitute the predominant error type.

The confusion matrices in Table 8 reveal that false positives unvoiced frames incorrectly labeled as voiced represent the dominant failure mode across all methods. Autocorrelation produces $4,861$ false positives, CREPE $4,238$, and YIN $3,313$. This asymmetry is consistent with Figure 8, where algorithms systematically fail to detect brief unvoiced intervals embedded between voiced regions.

Two distinct error patterns emerge:

1. **Temporal proximity errors**: When unvoiced intervals fall below approximately 100 ms, all algorithms fail to register the transition, treating the entire sequence as continuously voiced. This mirrors the pitch estimation limitation identified in Section 4.1 and stems from the same window-contamination mechanism: the analysis window at the nominal unvoiced onset still contains voiced samples from the preceding frame, elevating periodicity metrics above threshold.

2. **Stochastic variability**: On sustained unvoiced segments, CREPE and YIN exhibit high variance in their voicing decisions (Figure 8), producing intermittent false positives despite the absence of periodic content. This reflects the inherent difficulty of threshold-based discrimination near the decision boundary, where small fluctuations in confidence scores or difference function minima trigger classification flips.

5 Conclusion

5.1 Summary of Findings

This report presented a comparative evaluation of three pitch estimation methods, Autocorrelation, YIN, and CREPE, across a range of acoustic conditions and two distinct tasks: fundamental frequency tracking and voiced/unvoiced classification.

On the pitch estimation task, CREPE consistently outperformed the two DSP baselines. In clean conditions it achieved near-perfect Raw Pitch Accuracy (99.20% at a 10-cent tolerance), and its variance remained approximately an order of magnitude lower than YIN and Autocorrelation across all noise types and SNR levels. Its confidence score further provides a natural and effective voicing indicator, yielding an F1-score of 0.818 on the TEN VAD dataset with a simple threshold rule. YIN offered a reliable middle ground: more stable than Autocorrelation for pitch tracking. Autocorrelation, despite its high variance on pitch estimates, produced the strongest voiced/unvoiced classification results (accuracy 83.75%, F1-score 0.8505), suggesting that its binary periodicity criterion is well suited to the detection task even when its continuous frequency output is unreliable.

Three consistent failure modes were identified across all methods. First, **transition errors**: when voiced and unvoiced regions are separated by less than approximately 100 ms, window contamination prevents any of the three algorithms from resolving the boundary, producing spurious pitch activity. Second, **structured noise**: the presence of a competing harmonic source such as the trumpet interference used in Section 3 caused a catastrophic breakdown in CREPE (MSE $>$ 110,000 at 10 dB), because the model cannot separate two simultaneous pitched sources, a limitation shared in principle by all three methods since they are designed for monophonic signals. Third, **stochastic inputs**: CREPE exhibits erratic behaviour on purely aperiodic frames due to the distribution shift between its clean-speech training data and noisy inference conditions, while Autocorrelation collapses to extreme variance under the same conditions.

5.2 Future Work

Several directions could address the limitations identified above. The most direct improvement to CREPE would be the introduction of a **temporal smoothness constraint** during training for instance, adding a regularisation term penalising large frame-to-frame pitch jumps exploiting the well-established physical property that f_0 evolves continuously in natural speech and music. This would reduce the erratic estimates observed during noisy unvoiced frames without requiring architectural changes.

For voiced/unvoiced classification specifically, the results suggest that combining the complementary strengths of the three methods, CREPE's low false-negative rate, YIN's low false-positive rate, and Autocorrelation's overall F1 performance, through an ensemble or learned fusion could yield gains beyond any single method.

More broadly, extending the framework to **multi-pitch estimation** would remove the single-source assumption that underlies all three methods and is responsible for the catastrophic failure under trumpet interference. Recent neural approaches such as **YinNet** and **SwiftF0** have already demonstrated improvements over CREPE in both noise robustness and computational efficiency for the monophonic case, and represent a natural starting point for such an extension.

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